

OSINT Social Media Forensics Investigation System Report

1. Abstract

This report presents an advanced Open Source Intelligence (OSINT) system designed to collect, analyze, and visualize publicly available data across multiple social media platforms. The system integrates APIs, web scraping, browser automation, and an interactive dashboard to provide a unified digital forensic investigation tool.

The main objective of this project is to demonstrate how OSINT techniques can be applied in cybersecurity investigations to track digital identities, detect behavioral changes, and correlate user activity across platforms such as GitHub, Reddit, Twitter, and Instagram. The system also includes forensic snapshot analysis, change detection mechanisms, and social graph visualization.

By combining multiple data sources, the system provides a comprehensive view of a target's digital footprint while maintaining modularity and scalability for future improvements.

2. Introduction

Open Source Intelligence (OSINT) refers to the process of collecting and analyzing publicly available information from online sources. In cybersecurity and digital forensics, OSINT is widely used for threat intelligence, identity tracking, investigation support, and behavioral profiling.

In modern digital environments, individuals often leave traces across multiple platforms such as GitHub, Reddit, Instagram, and Twitter. Manually analyzing this data is inefficient and time-consuming. Therefore, automated systems are required to streamline the investigation process.

This project introduces an OSINT framework that automates data extraction, comparison, and visualization. It is designed to help investigators identify changes in user behavior, detect suspicious activity, and reconstruct digital identities.

3. System Overview

The system is composed of multiple integrated modules:

- Cross-platform username verification
- GitHub API-based data extraction
- Reddit profile analysis using JSON API
- Instagram scraping using Selenium automation
- Snapshot storage system for forensic comparison
- Change detection engine for behavioral analysis
- Social graph visualization using PyVis
- Streamlit-based interactive dashboard
- Wayback Machine integration for deleted content recovery

Each module operates independently but contributes to a unified OSINT investigation pipeline.

4. Cross-Platform Username Enumeration

This module checks whether a username exists across multiple platforms using HTTP requests. It targets APIs from GitHub, Twitter, and Reddit.

The system interprets HTTP status codes as follows:

- 200 → User exists
- 404 → User not found
- 401 → Unauthorized access
- 429 → Rate limit exceeded

This allows investigators to quickly determine where a digital identity exists across platforms. It is a fundamental step in OSINT reconnaissance.

5. Instagram Data Extraction (Selenium Automation)

Instagram does not provide a fully open API for profile scraping, so Selenium is used to automate browser behavior.

The system performs the following steps:

- Opens Instagram login page
- Automatically enters credentials (if required)
- Navigates to a target profile
- Extracts username, follower count, following count, and post links

This approach simulates real user interaction, allowing access to dynamic content that is not available through static requests.

6. Reddit Profile Analysis

The Reddit module uses the public Reddit JSON API endpoint to retrieve user information.

Extracted data includes:

- Username
- Account creation date
- Total karma
- Profile image
- Verification status

The creation timestamp is converted into a human-readable format, which is useful for forensic timeline reconstruction.

7. GitHub Forensic Data Collection

GitHub is used as a primary data source due to its structured API and rich user metadata.

The system extracts:

- Username
- Bio
- Followers and following count
- Public repositories
- Location
- Account creation and update timestamps

This data is stored as snapshots, allowing historical tracking of user changes over time.

8. Snapshot Storage and Change Detection

The system saves user data in JSON format to maintain historical records.

A forensic comparison engine analyzes the last two snapshots and detects:

- Changes in followers or following count
- Bio modifications
- Location updates
- Repository changes

Each detected change is classified based on severity:

- LOW → Minor modification
- MEDIUM → Moderate activity change
- HIGH → Suspicious or large-scale change

This helps identify abnormal behavior patterns over time.

9. Social Graph Visualization

The system uses PyVis to generate an interactive social network graph.

Features include:

- Recursive follower mapping
- Node color classification:
 - Red → Target user
 - Orange → Followers
 - Blue → Secondary connections
- Directed edges showing relationships
- Physics-based layout for better visualization

This graph helps visualize digital influence and network structure.

10. Streamlit Dashboard

The system includes a Streamlit-based dashboard that provides an interactive user interface.

It includes multiple tabs:

- Profile Overview
- Social Graph
- Snapshot Change Analysis
- Cross-Platform Search
- Deleted Content Detection
- Final Investigation Report

The dashboard enables users to conduct investigations without needing technical expertise.

11. Deleted Content Forensics

This module compares historical snapshots to detect removed or modified data fields.

It can identify:

- Deleted bio information
- Disappearing location data
- Sudden follower changes
- Account modifications

Such changes may indicate privacy adjustments, account manipulation, or suspicious activity.

12. Wayback Machine Integration

The system integrates with the Internet Archive (Wayback Machine) to retrieve archived versions of profiles.

This allows investigators to:

- Recover deleted content
- Compare historical and current profiles
- Track long-term digital changes

It is especially useful for forensic investigations where content has been removed.

13. Security and Ethical Considerations

Although the system only uses publicly available data, ethical and legal considerations are critical.

Important guidelines include:

- Do not use the system for unauthorized surveillance
- Respect API rate limits and platform terms of service
- Protect API keys and authentication tokens
- Avoid scraping private or restricted data

The system is intended strictly for educational and research purposes.

14. Limitations

The system has several limitations:

- Social media APIs may change or restrict access
- Selenium scraping can break due to UI updates
- Rate limits may block frequent requests
- Large graphs may impact performance
- Twitter API access requires authentication and may be restricted

Despite these limitations, the system remains a strong OSINT foundation.

15. Conclusion

This OSINT forensic system demonstrates how multiple open-source tools and APIs can be combined into a powerful investigative platform. It successfully integrates data extraction, behavioral analysis, and visualization into a unified workflow.

The system provides insights into digital identities and their evolution over time, making it valuable for cybersecurity analysis, digital forensics, and research applications.

Future improvements may include machine learning-based anomaly detection, real-time monitoring, and enhanced correlation between platforms.